

Dynamic Geometry and Mathematics Education

Special Session at the 24th Conference on Applications of Computer
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Aim and scope

Dynamic geometry environments (DGE) have emerged in the last half-century with an ever-increasing impact in mathematics education. DGE enlarges the field of geometric objects subject to formal reasoning, for instance, simultaneous operations with many geometric objects. Today DGE open the possibility of investigating visually and formulating conjectures, comparing objects, discovering or proving rigorously properties over geometric constructions, and Euclidean elementary geometry is required to reason about them.

Along these decades various utilities have been added to these environments, such as the manipulation of algebraic equations of geometric objects or the automated proving and discovering, based on computer algebra algorithms, of elementary geometry statements. Moreover, some intelligent tutoring systems for Euclidean geometry based in DGE have been developed.

The merging of these tools (DGE, automated proving and intelligent tutoring systems) is, thus, a very natural, challenging and promising issue, currently involving logic, symbolic computation, software development, algebraic geometry and mathematics education experts all from over the world.

The Special Session intends to be a forum for:

- presenting the current state of the art concerning the design and implementation of automatic reasoning features on dynamic geometry systems and intelligent tutoring systems;
- fostering a debate concerning the role and use of such features in mathematics education, in general, and their potential impact in proof and proving conception in the classroom, in particular.

Organizers

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